

Heterogeneity in Informal Salaried Employment: Evidence from the Egyptian Labor Market Survey

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Summary. — This paper contributes to the continuing debates on the mechanisms driving labor market informality in developing countries by proposing an innovative way to discriminate between segmented and competitive markets. An empirical analysis is applied to Egyptian paid employment in the highly dynamic context of 1998–2006.

The study is based on recent nonparametric methods applied to estimate the model with essential heterogeneity. The model is extended to decomposing the treatment effects into unobserved and observed components.

The results show triple heterogeneity of workers on the Egyptian labor market, offering support to both segmented and competitive views on informal labor.

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Key words — North Africa, Egypt, labor market segmentation, informal employment, wage differentials, essential heterogeneity

1. INTRODUCTION

The correct identification of the nature of informal employment is crucial for the efficient application of labor regulations in the context of informal labor. No consensus has been achieved on the mechanism driving informal employment in developing countries when arguing in favor of either the dual market theory or comparative advantage analysis. The inconclusiveness of the debates is likely due to the complex structure of informal employment. This paper addresses the issue of the voluntary *vs.* involuntary nature of informal employment by applying the model with “essential heterogeneity”, a model allowing the detection of a correlation between workers’ gains from formal *vs.* informal labor allocation and the allocation process.

The dual market theory assumes that there are two distinct segments in the labor market: a formal segment paying high wages and providing job security and social security and an informal segment paying low wages and serving as a last resort to unemployment (Harris & Todaro, 1970). According to this approach, informal employment is a predominantly involuntary engagement of workers in a segmented labor market, the formal segment being rationed (Fields, 1990, 2005a; Harris & Todaro, 1970; Stiglitz, 1976).

An alternative approach states that informal employment results from the voluntary choice of workers and thus fits the competitive market framework (Maloney, 1999, 2004; Rosenzweig, 1988, chap. 15). It contrasts with the dual labor market theory, which ignores micro-level decisions such as an individual’s cost-benefit analysis¹. The competitive market view implies that informal workers might be better off with incomes earned informally than they would be with those earned formally. In regard to poor informal workers it implies that they would not be better off holding formal jobs for which they are qualified (Maloney, 2004). The opportunity for employment choice underlying the competitive market view might be exceedingly limited when the share of formal employment is low.

A number of attempts have been made to empirically test the relevance of segmentation. For example, the validity of the dual market theory was challenged by (Basu, 1997;

Cunningham & Maloney, 2002; Dickens & Lang, 1985; Heckman & Hotz, 1986; Magnac, 1991; Pratap & Quintin, 2006). The main idea of the tests is to look for grouping effects indicating the presence of two nonintegrated segments.

The results diverge, with some supporting each postulated extreme. For example, Dickens and Lang (1985) conclude that there is evidence of a dual labor market, finding different wage-setting mechanisms in the formal and informal segments and arguing that the returns to human capital should be the same in the context of an integrated market. Conversely, Maloney argues that conditional wage comparisons may be dubious measures given the different immeasurable, nonpecuniary benefits of formal and informal employment (Cunningham & Maloney, 2002). To take another example (Magnac, 1991) using similar approach, concludes that the market is competitive. However, his test is based on the implausible assumption that there is no correlation between preferences and observable or unobservable wage determinants except through monetary gains.

Overall, no strategy has been proposed that would allow researchers to discriminate between these two polar points of view. On the other hand, even the researchers working at the extremes of the two base theories admit that both theories are valid and operate simultaneously (Cunningham & Maloney, 2002; Fields, 1990).

The most recent theoretical studies suggest a combination of the two approaches, assuming that informal employment is heterogeneous (Fields, 2005b; Paulson & Townsend, 2005) and can consist of both voluntary and involuntary groups.

According to several empirical studies, the model of competitive, voluntary informal employment seems to be relevant to

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the most developed of developing countries (Maloney, 1999, 2004; Navarro-Lozano & Schrimpf, 2004). Additionally, most studies on informal employment validated the competitive market approach when studying informality in Latin American countries. Recent works on welfare analysis in these countries rely heavily on this literature and founds analysis on the assumption of a competitive labor market and not a dual labor market (see for example Meghir, Narita, and Robin (2012)).

Recent empirical studies on African and Asian labor markets instead put forward the idea of heterogeneity of informal employment. Recent literature offers a variety of approaches focusing on the demand side of the market (Böhme & Thiele, 2012; Grimm, Krüger, and Lay (2011)), food shortage (Floro & Swain, 2013) or intergenerational linkage and importance of family background of African workers in terms of informal employment and self-employment (Pasquier-Doumer, 2013).

Focusing on understanding informal employment, the present paper fits the research line followed by Gunther and Launov (2012) and Dimova, Nordman, and Roubaud (2010), who study African informal labor markets using the classical tools of selection and earnings equations.

Using household survey data from the Côte d'Ivoire, Gunther and Launov (2012) detect empirically several segments within the informal sector that differ in earnings, returns to human capital, and individual characteristics. They show that the distribution of workers among different segments is inefficient in terms of monetary earnings, which implies that some informal employment is involuntary and that informal activities are heterogeneous.

Dimova et al. (2010) also find differing returns to human capital and individual characteristics in the formal segment as compared to informal segments, defining the segments based on the self-employed *vs.* paid status of informal workers. They show various patterns of intra-sectorial selection, raising question of inefficient allocations in terms of unobserved skills as well as nonproductivity of the human capital in the informal sector.

This paper makes a three-fold contribution to the line of research that investigates the nature of informal employment in developing countries and is discussed above.

First, it proposes an innovative method of analysis which employs additional parameters and allows a more in-depth investigation of the behavioral patterns of informal workers. The application of the model with essential heterogeneity recently proposed by Heckman, Urzua, and Vytlačil (2006) allows the exploration not only differences of returns on observable human capital and unobservable skills of formal and informal workers, but also the exploration of individual formal and informal gaps in returns to unobservable characteristics. Essentially, it allows investigation of the distribution of the gaps and its relationship to the allocation process. In our context, the different types of the relationship are associated with different mechanisms driving informal employment.

Second, in contrast with the majority of studies analyzing earning equations, the present empirical analysis draws on Heckman et al. (2006) using nonparametric techniques. The estimation procedure relaxes the assumption that there is a specific shape of the distribution that the unobservable components of the model follow.

Finally, the paper offers an application to Egyptian informal paid employment, providing a case study of the labor market of one of the lower-middle-income African economies that is rather under-investigated. The data are drawn from 1998 and 2006 rounds of the ELMS household survey, allowing the comparison of results over a time period affected by

significant fluctuation in the Egyptian labor market, in particular the shrinking of the formal sector.

The key findings are in line with recent literature suggesting that informal employment has a complex structure. Different workers may have different relative gains from working formally *vs.* informally; the distribution of the gains across employment may be socially efficient or not depending on the nature of informal engagement. Formal and informal workers may have different levels of skill; informal workers may also differ in their skills and mechanisms inducing them to work informally; finally, different strata may follow different patterns in their allocation between formal and informal groups. Moreover, the Egyptian case study shows that the mechanism driving informality may evolve under labor market restructuring: the data show evidence of increasing segmentation during 1998–2006.

The paper is organized as follows. Section 2 describes the concepts of the treatment literature related to the model with essential heterogeneity and discusses how they apply in the context of informal employment. Section 3 presents the data and discusses the main features of the Egyptian labor market over the period of study. Section 4 presents the model. Section 5 discusses the main empirical results. Final remarks are summarized in the Conclusion.

2. FORMAL EMPLOYMENT AS A TREATED POPULATION

The output of the paper's empirical analysis is a set of "treatment" effects. In our framework we designate as treated the population of workers holding formal jobs and as untreated the population of workers holding informal jobs. Formal and informal earnings represent the two outcomes from being treated and untreated, respectively. A difference between formal and informal earnings for a given worker defines an individual treatment effect.

Most of treatment estimators developed in econometrics respond to the two major problems of evaluation: a missing counterfactual state and selection into treatment. In our framework, the first problem translates into observing the majority of workers as a part of either formal or informal employment and not both simultaneously. The second problem is the classical endogeneity of occupational choice: workers' engagement in formal *vs.* informal employment may have the same unobservable determinants as their earnings from formal *vs.* informal jobs.

An additional evaluation problem arises in the event of the "essential heterogeneity," when individual heterogeneity is also related to the treatment responses and their correlation with the assignment rule. In a policy evaluation context, this implies first that different agents benefit differently from the policy and second that agents are either subjected to the policy or not depending on individual gain or loss from whether or not they are subjected to the policy.

In our context, allowing for essential heterogeneity is particularly interesting. First, a difference between formal and informal wages for a given worker may be nonzero: there may be different returns to observable (human capital variables, gender) and unobservable (entrepreneurial ability, family background) determinants of formal and informal wages. Second, a nonzero correlation between individual gains or losses from holding a formal rather than an informal job and allocation between different types of employment would capture workers' eventual sorting based on the gain into formal *vs.* informal employment. On the other hand, a zero correlation would

signal that there is no cost-benefit analysis behind workers' allocation. The distribution of the individual treatment effect and its relationship with the assignment process may reveal behavioral variation driving the assignment into different groups. Positive sorting on the gain as opposed to zero sorting on the gain thus constitutes a discriminating condition between voluntary and involuntary informal employment. Section 4(c) specifies this in detail relating to the framework of the model described in Sections 4(a–b).

To date, there are a limited number of applications of the model with essential heterogeneity. Most of them are applied to studying the returns to schooling (Carneiro, Heckman, & Vytlacil, 2011; Carneiro, Lokshin, Ridao-Cano, & Umapathi, 2011; Heckman & Li, 2004; Heckman et al., 2006; Reyes et al., 2013; Wang, Fleisher, Li, & Li, 2007) with the exception of the work of Basu, Heckman, Navarro-Lozano, and Urzua (2007) evaluating the costs of localized breast cancer treatments using the same approach.

A straightforward estimation of the model could be done using maximum likelihood estimator. This estimator would require specifying the joint distribution of unobservable determinants of the selection process and two outcomes (in our context formal or informal job assignment and formal or informal earnings). Most often, a multivariate normal distribution of the terms is hypothesized. However such a hypothesis does not have any economic foundation and most often is not verified empirically. Heckman, Tobias, and Vytlacil (2003) show that the bias resulting from a false law specification may be important. Heckman et al. (2006) suggest a method of nonparametric estimation, relaxing this assumption in regard to the treated and untreated outcomes (formal and informal wages in our framework). They apply the procedure to the question of the return to education.

The specification in our context, as compared to the return to education applications, is characterized by unequal distribution of the observable explanatory variables among the two different groups, informal and formal employees, as well as the discrete nature of these variables. Thus, we decompose the treatment effect into unobserved and observed components and adjust the procedure for multiple discrete control variables.

From the perspective of policy implications, the analysis highlights the efficiency of the allocation of workers between formal and informal segments in terms of social welfare: it shows whether workers' gains from holding one kind of job rather than another are maximized. The efficiency of the allocation is an important dimension of the impacts of policies targeting the formalization of employment: such policies may increase job security on the one hand but decrease social welfare on the other hand.

Next, the analysis is instructive for policies targeting an increase of workers' social protection and/or tax collection by reducing informal employment and thus drawing workers into formal employment. The goal of such policies is to reallocate workers. Understanding the nature of the allocation process is crucial here.

The labor supply in markets in which cost-benefit considerations prevail may respond positively to formalization incentives which improve formal social packages and introduce a formal minimum wage. Conversely, a market with a segmented labor supply may have no response to such policies. In the latter case, the incentive mechanisms cannot be related to monetary or nonmonetary gains, but should focus on tackling the barriers to entry into the formal sector through increased formal sector job creation, increased flexibility in

regulation of the hiring process, and legislative and tax regulations in favor of business startups.

Finally, the analysis is instructive about the distribution of workers' returns to observable and unobservable individual characteristics. The above-mentioned policies may therefore have disparate impacts on the various employment strata defined by both observed and unobserved characteristics.

3. EGYPTIAN DATA AND MARKET EVOLUTION

The data we use are drawn from the household survey administrated by the Economic Research Forum (ERF) in cooperation with the Egyptian Central Agency for Public Mobilization and Statistics (CAPMAS)—the main statistical agency of the Egyptian government. The survey tracks the labor market and demographic characteristics of households and individuals. The data set is nationally representative and consists of two rounds: ELMS 98 and ELMS 06 allowing comparisons during 1998–2006.²

The Egyptian labor market experienced significant fluctuation during 1998–2006. Assaad (2009) outlines several sources of the market's evolution over this period. First, before 1998, employment in the government sector was rapidly growing in spite of the beginning of a movement to reduce employment in state-owned enterprises. From 1998 to 2006, employment growth in the civil service decreased drastically, leaving the private sector to absorb most of the increase in the labor force. Also, demographic changes contributed to the growth of the labor force over this period: significant decreases in early childhood mortality in the 1980s preceded with high fertility rates caused large population growth. The peak of the population of 22-year-olds was observed in 2006. Additionally, the educational composition of the labor force was characterized by a significant increase of the fraction of technical secondary school graduates balanced by reduction in the proportion of males who are illiterate. Finally, new labor legislation was implemented in 2003 with the goal of liberalizing the labor market. The new law introduced flexibility into formal employment relationships, easing the process of hiring and firing as well as allowing fixed-term contracts.

The increasing labor force was well-absorbed into the private sector: according to Assaad (2009), unemployment has declined from 11.7% in 1998 to 8.3% in 2006. However, the growth in private formal employment has not been sufficient to absorb both the growing labor force and the inflow of large cohorts of civil servants and young workers. Consequently, informal employment has grown significantly.

The restructured labor market is likely to operate differently in 2006 as compared to 1998. The nature of job assignment among formal and informal employments could go different ways. Indeed, the growing labor force should have exerted strong supply pressure on the labor market. At first sight it seems as though the barriers to entry into the formal sector should have strengthened during 1998–2006. Stronger barriers to entry may reinforce the last-resort quality of informal employment and favor segmentation. On the other hand, the new legislation and more advanced educational composition of workers might be favorable for employment formalization and competitiveness. Given such multifaceted dynamics, the analysis of data is imperative.

The 1998 sample (ELMS 98) was drawn from a master sample of 750,000 households prepared by CAPMAS. It covers 4,816 households. The 2006 sample (ELMS 06) is made up of three different groups. The first group is 3,684 households from the original 1998 survey. It makes 44% of the final

number of the households in 2006. An analysis of the attrition showed that it was essentially due to the random loss of identifying records. Weights based on the probability of nonresponse were used by the data administrators to correct for attrition (Barsoum, 2007). The second group of 2,167 new households was formed as a result of splits from the original households. Given an important time span during 1998–2006, the 2006 sample could not remain nationally representative. Therefore, a third group of about 2,500 refreshing households was additionally drawn from the CAPMAS master sample to fill this gap and maintain a nationally representative sample. The refreshment was based on the same sampling procedure as the initial 1998 draw.³

The overall functioning of the labor market is the focus of this analysis. Thus, we are interested in representative samples of workers in both 1998 and 2006. With the significantly refresher sample, the analysis is within the frame of two repeated cross-sections.

Our sample is composed of individuals who are between 15 and 64 in age and engaged in the labor market.⁴ This provides us with a sample of 6,203 workers from the 1998 sample and 10,536 workers from the 2006 sample.

The sample distribution of employment by category based on status of employment is reported in Table 1. The present analysis focuses on the first category, the salaried employment: the data do not provide information on the earnings of employers and independent workers, but earnings is one of the key variables of the analysis. In addition, agricultural workers are not considered.⁵ While Cairo and Alexandria regions were over-sampled and rural areas under-sampled to increase the probability of obtaining women wage-workers in the private sector (Barsoum, 2007), the sample distribution adequately represents the population structure of employment (population distribution is given in parenthesis).

As it follows from Table 1, employees represent 77% of workers in 1998 and 72% in 2006. The corresponding sample yields 4,886 observations from 1998 and 7,609 observations from 2006. Eliminating observations with incomplete information on variables of interest yields the final sample of 4,176 and 6,847 employees observed in 1998 and 2006 respectively.

When defining informal employment, we rely on the job-based concept recommended by the International Labor Organization (ILO) denoting an informal job as an activity that is unregulated by the formal institutions and regulations of society. More specifically, this is a job characterized by an employment relationship that is “not subject to national labor legislation, income taxation, social protection or entitlement to certain employment benefits” (Husmanns, 2004). We then follow Assaad, 2009 by defining informal employment as employment that is not covered by either social insurance or a legal contract. The corresponding breakdown of the

employment does not necessarily correspond to employment in formal or informal sector firms. Since the CAPMAS policy was not to get data on the economic unit from the employee, it is impossible to observe specific affiliation to the formal or informal sector.

Representing more than 70% of the Egyptian nonagricultural employment, employees make up all of public employment and more than half of both private formal and informal employment.

As it is reported in Table 1, the share of public employment has decreased by about 8%, while the informal employment share has increased by about 9%. The employees’ population follows a similar pattern: an 11% decrease in the public employment share and a 9% increase in the informal employment share. The data imply that the growth in private formal employment has not been sufficient to absorb the growing labor force and inflow of large cohorts of civil servants and young workers from 1998 to 2006 as evidenced by the significant growth in informal employment over this period.

(a) Formal vs. informal employment: composition

Formal and informal employment are composed of different economic activities. The economic activities that are basically informal are construction, trade, and hotel and restaurant services (Table 2 reports the population distribution). These sectors were the most dynamic from 1998 to 2006, growing faster than overall employment (Assaad, 2009). Essentially formal activities are public and health services, education, finances and utilities. A substantial portion of both types of workers is employed in manufacturing (29% of informal employment in 1998 and 23% in 2006; 16% of formal employment in 1998 and 15% in 2006).

In terms of economic activities, the formal employment structure is quite stable from 1998 to 2006 (Table 2). The informal activities have been slightly redistributed: the percentage of workers employed in manufacturing and construction has slightly decreased, while the percentage of trade and communications employees has slightly increased.

Table 3 reports the sample distribution of the discrete explanatory variables across the formal and informal groups of employees.

In terms of the treatment literature, formal and informal populations represent treated and untreated groups. A standard requirement of any treatment evaluation estimator is to insure comparability of the two populations (formal and informal): for each treated agent with observable characteristics X , (such as age, level of education, etc.), there should be at least one untreated agent in the sample with the same characteristics. Formally, the requirement translates as a nonzero probability of being treated (propensity score) and untreated for

Table 1. ^aStructure of nonagricultural employment (family workers with no wage are excluded)

	Employment		Public		Private formal		Private informal	
	1998	2006	1998	2006	1998	2006	1998	2006
Employees, %	79 (77)	72 (72)	100	100	51 (51)	60 (62)	68 (67)	57 (57)
Employers, %	11 (12)	16 (16)	–	–	30 (30)	26 (25)	15 (16)	24 (25)
Self-employed, %	10 (11)	14 (14)	–	–	19 (19)	14 (13)	17 (17)	19 (20)
Total number	6,203	10,536	2,620	3,619	994	1,605	2,589	5,312
Total, %	100	100	42 (38)	34 (30)	16 (15)	15 (14)	42 (47)	51 (56)
Employees	4,886	7,609	2,620	3,619	506	963	1,760	3,027
Employees, %	100	100	54 (56)	47 (45)	10 (11)	13 (13)	36 (33)	40 (42)

Source: ELMS 98 and ELMPS 06.

^a The population distribution is given in parenthesis.

Table 2. *The structure of employment by sector of economy, population distribution*

	1998		2006	
	Informal frequency	Formal frequency	Informal frequency	Formal frequency
Mining	12,355 (0.4%)	22,709 (0.3%)	5,734 (0.1%)	51,276 (0.7%)
Manufacturing	934,217 (29%)	1,089,161 (16%)	1,133,696 (23%)	1,141,014 (15%)
Electricity, gas, water	4,094 (0.1%)	143,924 (2%)	7,146 (0.1%)	185,753 (2.4%)
Construction	867,343 (27%)	185,035 (3%)	1,149,293 (24%)	226,719 (3%)
Trade	464,905 (14%)	238,559 (4%)	1,009,450 (21%)	317,229 (4%)
Hotels and restaurants	136,260 (4%)	68,921 (1%)	313,946 (6%)	118,012 (1.5%)
Transport and communications	193,389 (6%)	569,912 (9%)	489,104 (10%)	753,284 (10%)
Finances	9,449 (0.3%)	151,003 (2%)	13,206 (0.3%)	208,981 (3%)
Real estate and business	49,536 (1.5%)	53,651 (0.8%)	127,314 (3%)	114,603 (1.5%)
Education	59,898 (1.8%)	1,772,650 (25.5%)	86,570 (2%)	1,799,010 (23%)
Public Services	97,887 (3%)	1,683,474 (26%)	177,792 (4%)	1,937,345 (25%)
Health and health services	38,578 (1%)	394,278 (6%)	77,688 (1.6%)	567,866 (7%)
Social and community services	372,314 (12%)	103 (4%)	255,910 (5%)	291,824 (4%)
Total	3,240,225	6,614,423	4,846,849	7,712,916

Source: ELMS 98 and ELMP5 06.

workers with the same observable characteristics (Heckman et al. (2006)). This common support requirement is ensured by sampling formal and informal workers whose propensity scores (probability of working in formal sector) overlap. The identification of the treatment effects underlying empirical analysis of the paper and discussed in Section 4 relies on two properties of the common support: its strength (overlapping intensity) and fullness (closeness of the region of the common support to $[0;1]$ interval).

We construct the common support of the propensity distributions as $(\max(p_{0,\min}, p_{1,\min}); \min(p_{0,\max}, p_{1,\max}))$, where $p_{0,\min}$, $p_{0,\max}$ and $p_{1,\min}$, $p_{1,\max}$ are the minimal and maximal values of propensity scores for the samples of informal and formal workers. The distributions of the scores in the samples of the formal and informal employees in 1998 and 2006 are shown respectively by Figures 5 and 6. As can be seen from Table 3, the resulting common support is almost full: only few observations outside of the common support have to be dropped out. The empirical results therefore apply to the whole population described above.

The majority of highly educated workers hold formal jobs while the majority of workers with little or no education hold informal jobs. Despite the abolishment of the guarantee that all college graduates would have employment by 1982, the education level of the population has continued to increase over the last two decades (Assaad, 2009). This is confirmed by the educational distribution among workers from 1998 to 2006 (Table 3). As can be seen in Table 3, women's participation in the Egyptian labor market remained low from 1998 to 2006 with the majority of Egyptian women holding formal jobs. This is in line with finding of Assaad (2005) which draws attention to the defeminization of the Egyptian private sector

and female attachment to the governmental sector over the period of study.

Population distribution by formal and informal employment also varies by age and experience level. The youngest workers (15–24 years old) are usually informal employees (about 80% over the whole period). More than a half (60%) of the oldest workers (50–64 years old) are formal employees. The distribution of workers of 25–49 years old has been less stable during 1998–2006: While roughly the same percentage takes part in private formal employment in 1998 and 2006, workers are redistributed between informal and public formal employment with a significant decline in public employment (12%) and growth of informal employment (10%). The population studied in the present work, the employee population, shows a very similar cohort distribution with even sharper design. Thus, informal salaried jobs may offer young workers an entry point to the labor market, allowing them to compensate for a lack of schooling or skills.

Table 4 reports sample distribution of the continuous variables used in the analysis.

(b) *Formal/informal employment: wage differentials*

Figures 1–4 show the wage distributions (in logarithm forms) of the formal and informal employees in 1998 and 2006 respectively.⁶

Figures 1 and 2, based on the whole sample, show that in each period the wage distribution among formal employees is statistically superior to that of informal employees.

The wage differential might reflect various phenomena. First of all, it may be due to the heterogeneity of individuals from different groups. As it is shown in the previous subsection,

Table 3. *Distribution of the discrete variables by the formal and informal groups of employees over the common support*

	1998		2006	
	Informal	Formal	Informal	Formal
<i>Level of education</i>				
Nothing	457	418	652	459
Primary	224	233	379	273
Preparatory	101	145	166	172
Intermediate	300	926	984	1,550
Tertiary/University	105	1,093	394	1,793
<i>Gender</i>				
Male	1,073	2,167	2,223	3,122
Female	114	648	352	1,125
<i>Marital status</i>				
Yes	497	2,202	1,210	3,513
No	690	613	1,365	734
<i>Another member of the household working in the formal sector</i>				
Yes	367	1,354	668	1,818
No	820	1,461	1,907	2,429
<i>Region of residence</i>				
Greater Cairo	276	665	424	766
Alexandria and Suez Canal Cities	131	406	303	610
Urban Lower Egypt	205	480	364	597
Urban Upper Egypt	178	577	432	900
Rural Lower Egypt	236	448	576	917
Rural Upper Egypt	161	239	476	457
Common Support	1,187	2,815	2,575	4,247
Whole Sample	1,188	2,988	2,577	4,270
<i>Total number of observations</i>				
Common Support		4,002		6,822
Whole Sample		4,176		6,847

Source: ELMS 98 and ELMPS 06.

Table 4. *Distribution of the continuous variables by the formal and informal groups of employees over the common support*

	1998		2006	
	Informal Mean (Std-Dev)	Formal Mean (Std-Dev)	Informal Mean (Std-Dev)	Formal Mean (Std-Dev)
<i>Workers with any level of education</i>				
Ln(hourly wage)	0.425 (0.668)	0.691 (0.623)	0.516 (0.707)	1.050 (0.825)
Experience	12 (10)	20 (12)	13 (11)	19 (12)
Age	29 (9)	40 (10)	28 (10)	39 (10)
<i>Workers without university education</i>				
Ln(hourly wage)	0.421 (0.657)	0.551 (0.586)	0.520 (0.693)	0.907 (0.831)

Source: ELMS 98 and ELMPS 06.

formal and informal employees differ in their individual characteristics. Indeed, the superiority of the formal wage distribution over the informal wage distribution is weaker when considering employees without university education (Figure 3 vs. Figure 1 and Figure 4 vs. Figure 2).

Workers may also have different unobserved characteristics. On the one hand, observed or unobserved heterogeneity may be due to the labor market segmentation and thus selection from the demand side of the market. On the other hand, it may be due to the occupational choice of individuals based on their anticipated earnings.

4. MODEL WITH ESSENTIAL HETEROGENEITY: FORMAL AND INFORMAL WAGE GAP FRAMEWORK

(a) *Model*

The starting point of the model with essential heterogeneity is Roy's model of occupational choice (Roy, 1951). We consider working individuals i employed either in the formal ($D_i = 1$) or informal ($D_i = 0$) employment groups. The breakdown into two groups is very general here: depending on the

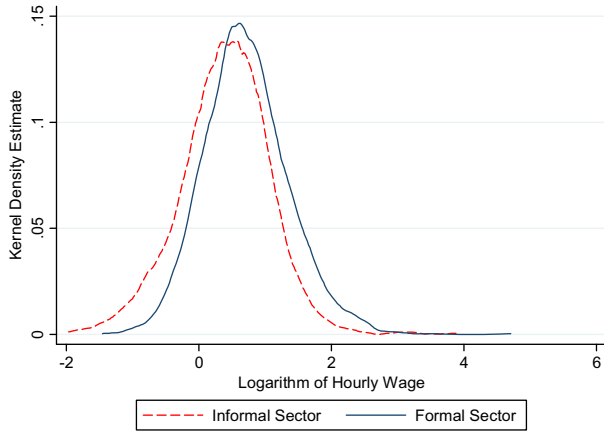


Figure 1. Wage distribution among employees with any level of education, 1998.

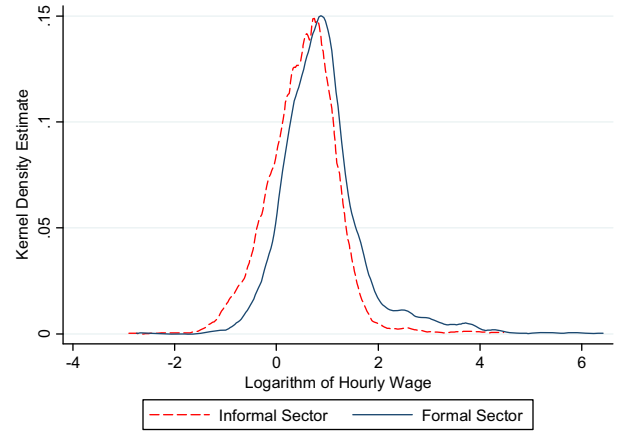


Figure 4. Wage distribution among employees without tertiary education, 2006.

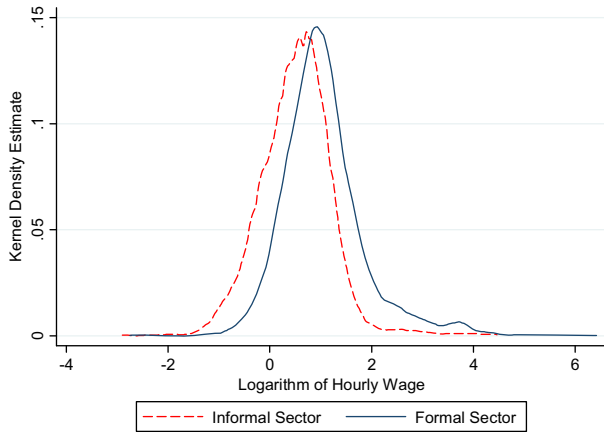


Figure 2. Wage distribution among employees with any level of education, 2006.

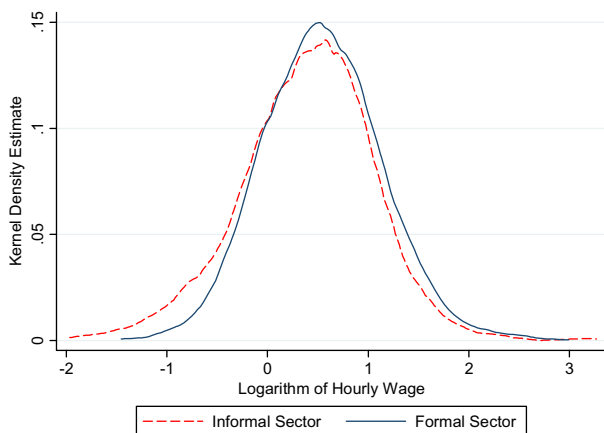


Figure 3. Wage distribution among employees without tertiary education, 1998.

context, it may be done based on enterprise-related characteristics representing formal *vs.* informal sectors or job-related characteristics representing formal *vs.* informal employment as it is in the empirical application of this paper. The observed

and unobserved heterogeneity is modeled by allowing two potential outcomes (i.e., formal *vs.* informal wage $W_{1,i}$, $W_{0,i}$) to vary with observed individual characteristics X_i and unobserved $U_{0,i}$, $U_{1,i}$:

$$\begin{cases} \ln W_{1,i} = \alpha_1 + \beta_1 X_i + U_{1,i}, & \text{if } D_i = 1 \\ \ln W_{0,i} = \alpha_0 + \beta_0 X_i + U_{0,i}, & \text{if } D_i = 0 \end{cases} \quad (1)$$

where α_0 , α_1 , β_0 , β_1 are unknown parameters of the wage equations. $E(U_{0,i}|X_i) = 0$, $E(U_{1,i}|X_i) = 0$.

System (1) is equivalent to the following wage Eq. (2):

$$\ln W_i = \alpha_0 + \beta_0 X_i + \Delta \cdot D_i + U_{0,i} \quad (2)$$

where individual “treatment” effect Δ is:

$$\Delta_i = (\alpha_1 - \alpha_0) + (\beta_1 - \beta_0)X_i + (U_{1,i} - U_{0,i}) \quad (3)$$

Thus, in addition to homogenous wage gap $(\alpha_1 - \alpha_0)$, the treatment effect takes into account wage differentials related to observable individual characteristics ($ATE(x_i) = (\beta_1 - \beta_0)X_i$) and unobserved treatment heterogeneity $(U_{1,i} - U_{0,i})$. $(U_{1,i} - U_{0,i})$ describes essential heterogeneity, heterogeneous difference of one’s returns from unobservable wage determinants earned if holding job 1 rather than job 0.

The participation equation is defined by

$$D_i = \begin{cases} 1, & \text{if } I_i > 0 \\ 0, & \text{if } I_i \leq 0 \end{cases} \quad \text{with } I_i = \gamma Z_i - V_i \quad (4)$$

where I_i is a latent variable representing benefit function (possibly negative) from working in the formal rather than informal segment, given observed and unobserved characteristics Z_i and V_i . Unobservables $U_{1,i}$, $U_{0,i}$, and V_i are correlated.

(b) Treatment effects and selection process

System (2) – (4) allows the definition of a set of parameters describing the distribution of the treatment effects and its relationship with the selection process.

The average treatment effect, ATE , represents the average formal/informal wage gap in our framework: $ATE = E(\ln W_1 - \ln W_0)$. If Δ_i and $U_{0,i}$ are uncorrelated with D_i , the population ATE can be estimated by the OLS method. Otherwise, correlation between (3) and (4) induces bias of the OLS estimator (Heckman & Vytlacil, 2007, chap. 71):

$$E(U_{1,i}|D_i = 1) - E(U_{0,i}|D_i = 0) = SG_1^U + SB_{1 \rightarrow 0},$$

where $SG_1^U = E(U_{1,i} - U_{0,i} | D_i = 1)$ denotes Sorting on the Gain and $SB_{1 \rightarrow 0} = [E(U_{0,i} | D_i = 1) - E(U_{0,i} | D_i = 0)]$ denotes the Selection term.

Sorting on the gain comes from one's anticipation of the sectorial wage differential given one's unobservable characteristics when making an occupation choice. The sorting on the gain is not zero unless $U_{1i} = U_{0i}$ or $(U_{1i} - U_{0i}) \perp D_i$. These cases are nested by the model with essential heterogeneity. The implications of both, zero sorting and non zero sorting options allowed by the model are discussed in Section (c).

The Selection term refers to the classical selection bias capturing the gap related to unobservable heterogeneity of workers employed in different groups.

In the event that the return to unobservable characteristics is not the same in different employment groups ($U_{1i} \neq U_{0i}$) and, that assignment to different groups is selective ($E[(U_{1i} - U_{0i}) | D_i] \neq 0$), the average treatment effect conditional on a given level of observables X is not the same among the populations of the two groups and is different from ATE . The treatment effect (return to working formally rather than informally) for the formal workers for worker, ($TT(x)$), is defined then as:

$$TT(x) = E(\ln W_1 - \ln W_0 | X = x, D = 1) \\ = ATE(x) + SG_1^U$$

where $ATE(x) = E(\ln W_1 - \ln W_0 | X = x)$. $TT(x)$ higher than $ATE(x)$ implies positive sorting on the gain ($SG_1^U > 0$): a worker choses job 1 expecting higher return to unobservable wage determinants from this job.

The same gap within informal population, ($TNT(x)$), is defined similarly:

$$TNT(x) = ATE(x) + SG_0^U$$

where $SG_0^U = E(U_{1i} - U_{0i} | D_i = 0)$. $TNT(x)$ lower than $ATE(x)$ implies positive selection on the gain into informal employment.

According to the descriptive analysis of Section 2, a specificity of the formal/informal job allocation is unequal distribution of the observable productivity characteristics among formal and informal workers ($E(X|D) \neq E(X)$). We extend thus the concept of the sorting on the gain introduced by Heckman et al. (2006), to relate it to both, observable and unobservable factors. We decompose the total (unconditional on X) sorting on the gain, SG_1 and SG_0 , in SG_j^U ($j = 0, 1$) and $SG_j^X = (\beta_1 - \beta_0)(E(X|D = j) - E(X))$ coming from observable human capital characteristics:

$$SG_1 = TT - ATE = SG_1^X + SG_1^U$$

$$SG_0 = TNT - ATE = SG_0^X + SG_0^U$$

Finally, a series of local treatment effects completes the set of parameters used as the tools of analysis. More specifically, marginal treatment effect, MTE , is the local average treatment effect measuring the change in aggregate outcome due to an infinitesimal change in the participation rate. The corresponding estimand is given by

$$MTE(X_i, U_{D,i}) = \frac{\partial E(\ln W_i | X_i = x_i, P(Z_i) = p)}{\partial p} \Big|_{p=U_{D,i}} \\ = E(\ln W_{1,i} - \ln W_{0,i} | X_i = x_i, U_{D,i} = u_{D,i})$$

where $U_{D,i}$ is the uniform transformation of the selection equation error term V_i , normalizing its range of values to $[0,1]$ using the normal cumulative density function F_V : $U_D = F_V(V)$.

In the framework of model (2)–(4), MTE is additive in its components related to observables (X) and unobservables (U_D) (Appendix A provides more details):

$$MTE(x_i, u_{D,i}) = \Delta\beta \cdot x_i + MTE_i^U$$

where $\Delta\beta = \beta_1 - \beta_0$ and MTE_i^U is the is the marginal gap between worker's i formal and informal earnings resulting from infinitesimal change of $U_{D,i}$ (or V_i since $U_{D,i}$ is its monotonous transformation). MTE is calculated at the point where worker i is indifferent about working in segment 1 or segment 0. At this point $V_i = \gamma Z_i$ which is identical to $F_V(\gamma Z_i) = F_V(V_i)$. The latter provides a set of $U_{D,i}$ defined in terms of propensity score $P(Z)$ to work formally: $U_{D,i} = P(Z_i)$.

MTE is thus the marginal gap between one's formal and informal earnings resulting from infinitesimal change of the participation rate.

Average treatment effects ATE , TT , and TNT can be recovered by integrating MTE over the distribution of the probability of participation implying calculus of the MTE weighted averages.

The estimation of parameters requires several steps that are detailed in Appendix A. When outlining the procedure, we specify the estimation particularities adopted given the data set in use.

(c) Model implications

The model described above provides a set of parameters to analyze. A subset of the parameters is the marginal treatment effect, MTE^U . MTE^U are V -specific, i.e., $P(Z)$ -specific. They are calculated at different points of the common support of the distributions of the propensity scores of the formal and informal populations. While the effect is not individual-specific, it is spread over the whole population of workers in case of the full common support or over a fraction of workers corresponding to the common support otherwise. Behavior of MTE as a function of unobservables U_D is one of the central points of the present analysis.

High U_D values imply high V values and low $(\gamma Z - V)$. Thus, MTE estimated at such points are associated with workers more likely to work formally than informally based on their observable characteristics. Inversely, MTE calculated at low U_D values show minimum gain due to unobservable characteristics motivating formal employment rather than informal one.

Therefore, the positive (negative) MTE is a worker's marginal gain (willingness to pay) if working formally ($D = 1$) or marginal willingness to pay (marginal gain) if working informally ($D = 0$).

There may be three main MTE^U behavioral patterns: (1) MTE^U decreasing with U_D ; (2) MTE^U increasing with U_D and (3) flat MTE^U .

Realization of the first pattern would be a signal of comparative advantage considerations underlying allocation of workers in different employment groups, formal and informal. Indeed, MTE^U decreasing with U_D would imply that the allocation is positively correlated with the relative gains based on unobservable characteristics of workers. In the segmented labor market, such a correlation would not be possible: if workers could move to the formal segment offering a higher wage, the same wage for identical workers would prevail in both formal and informal segments of the economy. Thus MTE^U function decreasing with U_D indicates competitive rather than segmented mechanism driving the labor market. In this framework the average treatment effects corresponding to unobservable gaps within formal employment population

(TT_U), informal employment population (TNT_U) and total employment population (ATE_U) are ordered as $TT_U > ATE_U > TNT_U$. The latter implies a positive sorting on the gain: the average gain of the formal workers from holding formal rather than informal job, TT_U , is higher than the average gain across the whole population, ATE_U ; the average gain of the informal workers from holding formal rather than informal job, TNT_U , is the lowest in the population; in particular, a negative value of TNT_U , would imply positive gain of the informal workers from their actual allocation into informal employment.

The second pattern would imply negative correlation between relative gains based on unobservables and selection equation. This is an unlikely scenario but is one of the theoretical options allowed by the model. It would imply that the higher the relative gains from working formally due to unobservables, the weaker the preferences for this type of employment based on the same unobservables. Once again it would be a signal of the comparative advantage consideration rather than segmentation driving the labor market. An example fitting such an option could be preferences for entrepreneurial activity or job security. Let us imagine for instance that the return to entrepreneurial ability is higher from the informal employment due to tax evasion. However, a worker may want to choose the formal employment that provides job security. Inversely, let us imagine that the return to entrepreneurial activity is higher in the formal sector and the return is increasing with higher entrepreneurial ability. However, a worker may want to choose the informal job providing him or her with a higher degree of independence in entrepreneurial activity. The higher the worker's ability, the stronger may be his or her preferences for independence and the lower may be the preferences for security. In this framework, the total effects based on unobservables are ordered as $TT_U < ATE_U < TNT_U$ following negative sorting on the gain. The second pattern would imply therefore prevalence of nonpecuniary comparative advantages driving the allocation of workers between the two groups. However, this pattern should be considered with caution since it yields correlation between monetary and nonmonetary relative gains based on the same unobserved factors.

Finally, MTE^U orthogonal to U_D would imply no relationship between anticipated gains from occupational choice and workers' preferences. It could be the very special case where there is no relative gain due to unobservables: $U_{1,i} = U_{0,i}$. In that case $MTE^U = 0$ in the population. In a more general case however $MTE^U = \text{const}$ and $MTE^U \neq 0$, the relative gain is nonzero $U_{1,i} \neq U_{0,i}$. This pattern fits the segmented labor market: despite of heterogeneous individual relative gains from working formally or informally, the average formal-informal wage gap due to unobservables is the same across the whole population of workers ($TT_U = ATE_U = TNT_U$). Workers do not sort themselves based on the anticipated gains. In the framework of the presented model, this would be a signal of segmentation driving the allocation of workers between formal and informal employment.

In addition to the sorting on the gain from unobservables, we allow for sorting on the gain from observables considering TT_X , ATE_X , TNT_X (see Appendix A, point 6) for technical details). $TT_X > ATE_X > TNT_X$ would imply positive sorting on the gain from observables; $TT_X < ATE_X < TNT_X$ would imply negative sorting on the gain from observables. Finally, $TT_X = ATE_X = TNT_X$ would imply no sorting. In case of $TT_X = ATE_X = TNT_X = 0$ the latter pattern would imply both, equal returns to X from the formal and informal employment and equal distribution of X among formal and

informal workers. $TT_X = ATE_X = TNT_X > 0$ would imply higher return to X from the formal employment while equal distribution of X in the two groups.

Finally, the analysis is to be completed by considering the total effects TT , ATE , and TNT made up of both components, based on observables and unobservables (see Appendix A, (9)–(11)). The relationship between these effects traces the overall functioning of the market and shows the prevalence of the effects related to observable or unobservable factors.

5. RESULTS

(a) Selection model results

Following the procedure detailed in Appendix A, we first estimate the selection equation fitting the Probit model. When estimating the Probit model, we pool the data from both rounds (1998 and 2006) while allowing the effects of the explanatory variables to vary over time. Some of the effects are found to be stable: effects of primary/preparatory education, region of residency, and gender. Consequently, we do not allow their variation in the final specification used to predict the propensity scores (Table 5).

The distributions of the predicted propensity scores for formal and informal employees in 1998 and 2006 are shown by Figures 5 and 6 respectively.

In line with the descriptive statistics from Table 3, Table 5 shows that the level of education is an important determinant of the selection process. This result is common for developing countries and African economies in particular (see for example Dimova et al. (2010) reporting the results for the largest economic centers of Côte d'Ivoire, Mali, Benin, Senegal, Togo, Niger, and Burkina Faso). To strengthen the validity of our analysis, we run the estimations using different samples, specifically the whole sample of employees and the sample of those without a tertiary level of education. As shown by Figures 7 and 8, the latter population has a stronger common support and thus better satisfies the requirement of mixing two groups conditional on X as discussed in Section 3(a).

The logarithm form of the experience level is used in the empirical work. Given Egyptian data, such a specification fits the nonlinear impact of experience on wages better than the traditional quadratic form of the Mincer's equation.⁷ In line with the descriptive statistics discussed in Section 3, working experience increases the probability of holding formal job. The experience effect is slightly higher in 2006 than in 1998.

Contrary to the experience level, the effect of intermediate and higher education is lower in 2006 than in 1998. More educated workers became more likely to work informally in 2006. This result is reflected by Figure 7 vs. Figure 5 and Figure 8 vs. Figure 6. The common support is stronger in 2006 than in 1998. It is particularly strong when considering workers without university education.

In line with the descriptive statistics reported in Section 3, women are more likely to work formally over the whole period of study. Marital status as well is positively correlated with the probability of holding formal job. It may partly capture some individual behavioral propensity toward stability and legality. Regional difference in terms of employment allocation is not high. Among six main regions, only Alexandria & Suez Canal Cities region and Rural Lower Egypt employ more workers formally than does the region of Cairo (Greater Cairo), but the differences are not strong.

Finally, conditional on the explanatory variables, the probability of formal employment has decreased over the period

Table 5. *Selection equation: probit estimates*

Worker's demographic characteristic	Workers with any level of education Coefficient (Standard Error)	Workers without university education Coefficient (Standard Error)
Ln(experience)	0.567*** (0.031)	0.680*** (0.038)
Primary education	0.274*** (0.052)	0.301*** (0.053)
Preparatory education	0.620*** (0.066)	0.656*** (0.068)
Intermediate education	1.174*** (0.067)	1.275*** (0.071)
Tertiary/university education	1.948*** (0.080)	—
Greater Cairo (reference group)	—	—
Alexandria and Suez Canal Cities	0.205*** (0.052)	0.176*** (0.062)
Urban Lower Egypt	0.037 (0.051)	−0.057 (0.059)
Urban Upper Egypt	0.088* (0.048)	−0.008 (0.058)
Rural Lower Egypt	0.143*** (0.143)	0.107*** (0.054)
Rural Upper Egypt	−0.062 (0.052)	−0.125*** (0.059)
Female	0.488*** (0.043)	0.641*** (0.054)
Marital Status	0.642*** (0.037)	0.653*** (0.044)
Another household member engaged in the formal sector	0.407*** (0.034)	0.412*** (0.040)
<i>2006–1998 difference</i>		
Ln(experience)	0.084** (0.036)	0.033 (0.045)
Intermediate education	−0.214*** (0.077)	−0.274*** (0.081)
Tertiary/university education	−0.216*** (0.093)	—
2006 year dummy	−0.480*** (0.117)	−0.336*** (0.141)
Constant	−1.943*** (0.108)	−2.103*** (0.128)
<i>Number of observations</i>	11,023	7,446

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

under consideration. The last result is consistent with the context of the Egyptian labor market as discussed in Section 3: formal private employment has grown, but this growth was insufficient to absorb the reduction of the public sector and important demographic developments. The new legislation providing formalization incentives was not efficient to balance the inflow of civil and young workers in the private sector.

(b) *Formal vs. informal earnings*

The central results of the paper are the estimates of the parameters introduced in Section 4(b). This set of estimates describes the distribution of the earning gaps at different levels of the propensity scores (given by MTE) as well as the average gaps within different populations, such as formal paid employment (given by TT), informal paid employment (given by TNT) and total paid employment (given by ATE). The population average gaps ATE , TT , TNT are broken down in

TT_X , ATE_X , TNT_X related to the observables and TT_U , ATE_U , TNT_U related to the unobservables. The treatment effects are estimated for both the 1996 and 2006 populations. Table 6 reports results for each period (1998, 2006) and each sample (the whole sample of employees and the sample reduced to the employees without university education).

According to the presentation of Section 4, differences between ATE and TT/TNT imply two facts. First, the wage gap is heterogeneous: formal/informal wage gaps are individual-specific; second, the selection and wage equations are correlated: on average the formal/informal gaps are different within the three different populations of employees (formal, informal, or both considered together).

The 1998 results suggest a positive sorting on the gain (positive correlation between the gain and employment allocation). Indeed, the total effects are ordered as $TT > ATE > TNT$. The total gap between formal and informal wages is positive for the formal workers and negative for the informal ones. This

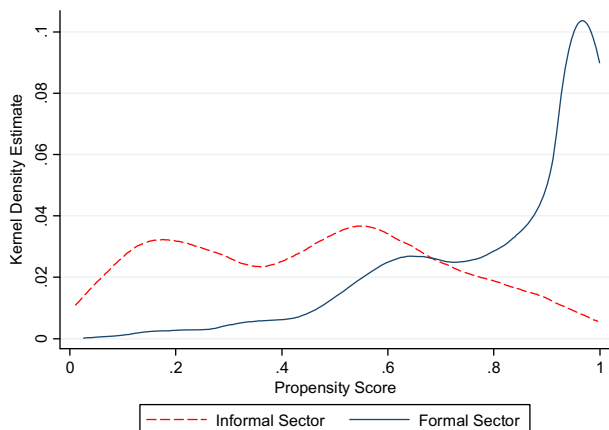


Figure 5. Distribution of propensity scores among employees with any level of education, 1998.

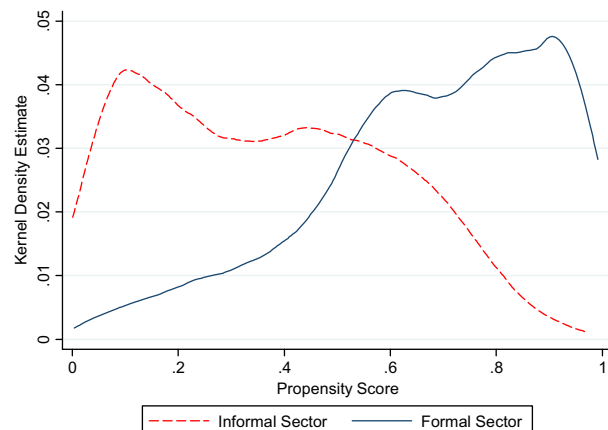


Figure 8. Distribution of propensity scores among employees without tertiary education, 1998.

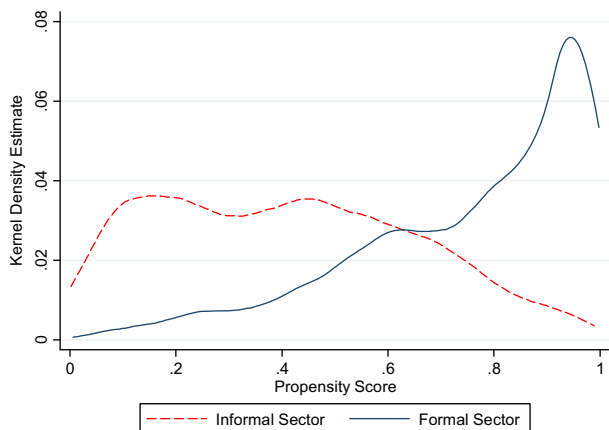


Figure 6. Distribution of propensity scores among employees with any level of education, 2008.

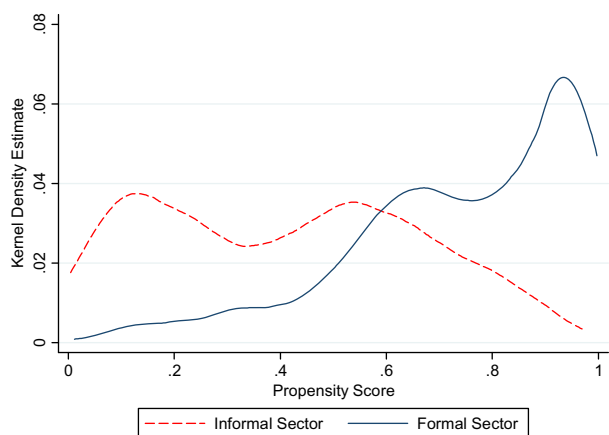


Figure 7. Distribution of propensity scores among employees without tertiary education, 1998.

implies positive difference between the actual wage and its counterfactual (the wage a worker would earn in case of different type of employment) for both, formal and informal workers. In general, both groups of employees benefit from their segment of employment more than they would do from the

other segment. The estimations carried out over the whole sample or reduced sample yield similar results: the average gain is about 50% for both types of workers except for the informal employees of the whole sample: they benefit slightly more (about 65%).

To better understand the estimated effects, we decompose formal-informal gaps into heterogeneous returns to observable (TT_X , ATE_X , TNT_X) and unobservable wage determinants (TT_U , ATE_U , TNT_U) (see Appendix A, point 6). Such a decomposition is relevant since on the one hand, the observable wage determinants such as experience and education (Tables 7 and 8) have different wage returns from formal or informal employment while being differently distributed across two employment groups. On the other hand, the unobservable wage determinants are differently distributed at different levels of propensity to work formally.⁸

According to the decomposition results, $TT_X > ATT_X > TNT_X > 0$ in 1998. The positive sign of the effects implies that the return to the observable characteristics is higher from formal than from informal employment for any group of employees: formal, informal, or the whole paid employment. However, the average gap is stronger among the formal workers than among the entire paid employment ($TT_X > ATT_X$), and it is the lowest among informal employees ($TNT_X < ATT_X$). Thus there has been strong self-selection of workers based on the expected gain from observable characteristics. The gain from the sorting is slightly stronger among the informal employees (about 30% *vs.* 15% among the formal employees) when considering the whole sample and about 20% in both groups when considering the employees without university education. Tables 7 and 8 report estimated coefficients of the wage equations obtained from the double residual semi-parametric regression detailed in section Appendix A (point 3).

In 1998, cumulated human capital was highly remunerated in the formal employment but not in the informal one. Indeed, the response of wage with respect to the experience is elastic (0.5) for the formal workers while is totally inelastic for the informal workers.

The return to education is as well very different in 1998: only tertiary education has an impact on wages in the informal employment (about 55%); in the formal employment the return to education increases progressively with the level of education (about 28%, 55%, 87%, and 110% higher wage when compared with the average wage of noneducated workers). At the same time, education is a strong determinant of the

Table 6. *Estimated Effects ATE, TT, TNT and their decompositions, 1998 and 2006*

Treatment effects	1998			2006		
	Total	TT_X ATE_X TNT_X	TT_U ATE_U TNT_U	Total	TT_X ATE_X TNT_X	TT_U ATE_U TNT_U
<i>Workers with any level of education</i>						
<i>TT</i>	0.251 ^{***} (0.076)	1.610	-1.359 ^{***}	0.510 ^{***} (0.122)	0.430	0.078
<i>ATE</i>	-0.345 ^{***} (0.088)	1.464	-1.810 ^{***}	0.554 ^{***} (0.100)	0.348	0.206 ^{***}
<i>TNT</i>	-0.999 ^{***} (0.119)	1.121	-2.120 ^{***}	0.518 ^{***} (0.153)	0.212	0.306 ^{***}
<i>SB_{1→0}</i>	1.026			-0.273		
<i>SB_{0→1}</i>	-1.787			0.499		
<i>Workers without university education</i>						
<i>TT</i>	0.291 ^{***} (0.106)	1.471	-1.179 ^{***}	0.555 ^{***} (0.122)	0.544	0.010
<i>ATE</i>	-0.229 ^{***} (0.084)	1.299	-1.529 ^{***}	0.311 ^{***} (0.115)	0.386	-0.074
<i>TNT</i>	-0.756 ^{***} (0.109)	1.060	-1.817 ^{***}	0.074 ^{***} (0.177)	0.217	-0.143
<i>SB_{1→0}</i>	0.970			-0.078		
<i>SB_{0→1}</i>	-1.608			-0.075		

Standard Errors are given in parenthesis.

*** Significant at 1%.

Table 7. *Wage equations: double residual semi-parametric regression estimates, workers with any level of education*

Worker's demographic characteristic	1998, Return from informal employment $\hat{\beta}_0$ (SE)	1998, Gap in formal-informal return $(\hat{\beta}_1 - \hat{\beta}_0)$ (SE)	2006, Return from informal employment $\hat{\beta}_0$ (SE)	2006, Gap in formal-informal return $(\hat{\beta}_1 - \hat{\beta}_0)$ (SE)
Ln(experience)	-0.002 (0.070)	0.493 ^{***} (0.088)	0.123 [*] (0.067)	0.180 ^{**} (0.091)
Primary education	-0.005 (0.084)	0.279 [*] (0.149)	-0.088 (0.084)	0.392 ^{**} (0.175)
Preparatory education	-0.117 (0.124)	0.552 ^{***} (0.194)	-0.004 (0.117)	0.353 [*] (0.207)
Intermediate education	-0.116 (0.159)	0.868 ^{***} (0.219)	-0.081 (0.118)	0.675 ^{***} (0.189)
Tertiary/university education	0.553 ^{**} (0.258)	0.546 [*] (0.322)	0.430 ^{**} (0.198)	0.284 (0.272)
Female	-0.340 ^{***} (0.107)	0.340 ^{***} (0.124)	-0.563 ^{***} (0.079)	0.636 ^{***} (0.104)
Greater Cairo (reference group)	-	-	-	-
Alexandria and Suez Canal Cities	-0.118 (0.031)	-	-0.053 (0.033)	-
Urban Lower Egypt	-0.142 ^{***} (0.028)	-	-0.152 ^{***} (0.032)	-
Urban Upper Egypt	-0.146 ^{***} (0.027)	-	-0.095 ^{***} (0.029)	-
Rural Lower Egypt	-0.191 ^{***} (0.029)	-	-0.147 ^{***} (0.030)	-
Rural Upper Egypt	-0.197 ^{***} (0.034)	-	-0.134 ^{***} (0.033)	-
Marital status	0.132 ^{***} (0.038)	-	0.111 ^{***} (0.041)	-
<i>Number of observations</i>	4,176	6,847		

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

Table 8. *Wage equations: double residual semi-parametric regression estimates, workers without university education*

Worker's demographic characteristic	1998, Return from informal employment $\hat{\beta}_0$ (SE)	1998, Gap in formal–informal return $(\hat{\beta}_1 - \hat{\beta}_0)$ (SE)	2006, Return from informal employment $\hat{\beta}_0$ (SE)	2006, Gap in formal–informal return $(\hat{\beta}_1 - \hat{\beta}_0)$ (SE)
Ln(experience)	0.076 (0.080)	0.468*** (0.114)	0.114 (0.079)	0.280** (0.128)
Primary education	0.045 (0.080)	0.229* (0.141)	−0.097 (0.083)	0.452*** (0.169)
Preparatory education	−0.001 (0.119)	0.423** (0.186)	−0.013 (0.119)	0.392** (0.208)
Intermediate education	0.112 (0.161)	0.606*** (0.227)	−0.025 (0.127)	0.633*** (0.208)
Female	−0.268*** (0.124)	0.249 (0.163)	−0.624*** (0.041)	0.900*** (0.149)
Greater Cairo (reference group)	—	—	—	—
Alexandria and Suez Canal Cities	0.018 (0.038)	—	0.007 (0.040)	—
Urban Lower Egypt	−0.073*** (0.035)	—	−0.088** (0.038)	—
Urban Upper Egypt	−0.130*** (0.035)	—	−0.059 (0.036)	—
Rural Lower Egypt	−0.129*** (0.035)	—	−0.066* (0.039)	—
Rural Upper Egypt	−0.165*** (0.039)	—	−0.070* (0.099)	—
Marital status	0.162*** (0.047)	—	0.120** (0.051)	—
<i>Number of observations</i>	2,804	4,638		

* Significant at the 10% level;

** Significant at the 5% level;

*** Significant at the 1% level.

selection equation (Table 5): workers with high education are more likely to be employed formally. Thus education simultaneously increases the probability to hold a formal job and the wage earned formally. It contributes therefore into the sorting on the gain from observables.

Gender is as well a factor of the sorting on the gain: women are more likely to hold formal jobs. At the same time, they earn less than men when working informally (about—34%) while there is no gender gap among formal employees when controlling for experience, education, and region of residency. The latter variable induces some wage variation (on the average, wages are lower in the regions other than Greater Cairo with variation about 12–20% and the lowest rate found in the Rural Egypt). The effects do not differ among different employment groups.

The results obtained when using the sample of employees without university education tell a very similar story (Table 8).

The results based on the whole sample also show positive sorting on the gain from unobservable characteristics $TT_U > ATT_U > TNT_U$ while the gap between formal and informal wages due to the unobservables is rather negative. SG_1^U and SG_0^U are about 45% and 30% respectively, implying that both types of workers gain on average more (or lose less in the case of formal workers) from their unobservables in their group of employment than they would earn in the other group of employment.

Again, the results based on the reduced sample are very similar with more equal sorting on the gain: about 30% in each sector.

The selection effects calculated using OLS estimation results reported in Tables 10 and 11 (Appendix B) are $SB_{1 \rightarrow 0} = 1$ and $SB_{0 \rightarrow 1} = -1.79$. The selection process is strong and asymmetric. The estimates imply that, should the formal workers

engage in the informal jobs, they would do twice as well as informal workers; conversely, should the informal workers engage in the formal employment, they would do more than twice as bad as the workers working there. Formal workers are found to have a higher performance than any other employment segment in 1998.

The average effects are calculated based on the marginal gains that are shown by Figures 9 and 10. The declining curves are consistent with $TT_U > ATT_U > TNT_U$ discussed above. Indeed, MTE decreasing with U_D implies that individuals less likely to hold formal job based on their observable characteristics (low U_D) and more likely to hold formal job based on the unobservable characteristics (low U_D corresponding to high $(-V)$ have the highest gain from the unobservables.

Overall, the results imply that the allocation of workers into formal and informal employment groups on the Egyptian labor market in 1998 is driven by the gains anticipated by employees from their observable and unobservable characteristics.

The 2006 results tell a different story. The overall average gaps between formal and informal wages are found to be roughly the same among different groups (formal, informal, and paid employment) of the 2006 sample. The overall treatment effect is homogenous, more specifically the formal employees are remunerated on the average 50% higher than informal employees. The homogenous positive treatment effect implies in our context the rationing of formal employment. Indeed, while any worker is better off from working formally, some workers are engaged in informal employment. Workers' behavior is not correlated with their relative gains. Therefore, the formal/informal assignment is not guided by workers' choice behavior but by restriction of the formal sector offering higher remuneration.

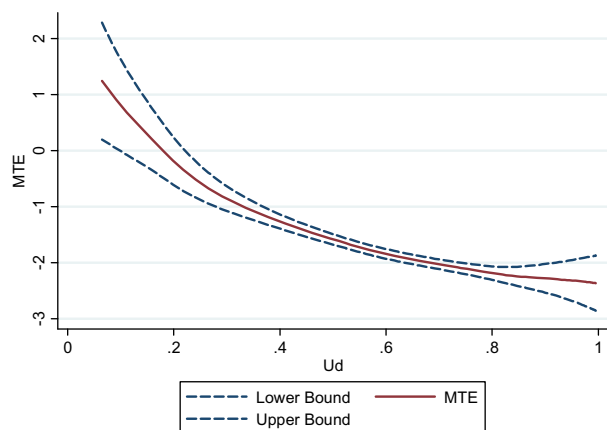


Figure 9. ^aMarginal treatment effect with 95% confidence interval bounds for employees with any level of education, 1998. ^aThe propensity score is ordered by the horizontal axis. Low abscissa values correspond to the workers who are less likely to be employed formally based on their observable characteristics. As discussed in Section 4(b), MTE are evaluated at the indifference points where the propensity score equals U_D . Low abscissa values are associated therefore with low V values and consequently high probability to be employed formally vs. informally based on unobservable characteristics holding the observables fixed.

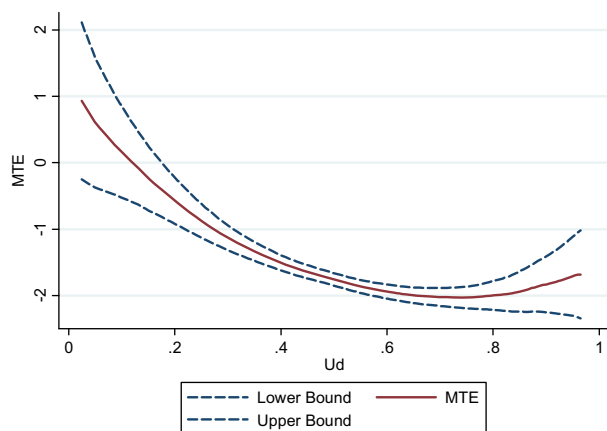


Figure 10. Marginal treatment effect with 95% confidence interval bounds for employees without tertiary education, 1998.

The decomposition of the effects into the gaps related to observables and unobservables show that in 2006 there is still positive sorting on the gain based on education and experience. It is, however, much less important than in 1998 and constitutes about 10% for both employment categories. It may be due to the reallocation of workers between the different groups and consequent changing return to the human capital: the wage response to experience becomes elastic in informal employment (0.12) while decreases to 0.18 in the formal employment. This difference from the 1998 pattern may imply increasing competition among informal employees having the same human capital.

There is a greater proportion of highly educated workers employed informally in 2006 than in 1998. The latter induces more equal distribution of education between formal and informal employment in 2006 as compared with 1998 and results in a less important wage gap related to the selection effect based on education. In addition, while the return to education in 2006 similarly to the return in 1998 is progressive

with the level of education in the formal employment, its magnitude is less important in 2006 than in 1998. Also, the difference in return to higher education in two groups decreases in 2006 (the difference coefficient is about 28% but statistically insignificant).

The positive sorting based on human capital is compensated by a negative sorting based on unobserved characteristics. However, the treatment effect related to unobservables is close to zero for formal workers while positive for informal ones. Figure 11 shows that MTE is basically negative, weak and flat except for the tails of the propensity score distribution. The tails corresponding to the workers having particularly high or low propensity to hold formal jobs contribute to nonzero treatment effects TNT_U , ATT_U , TT_U . Positive TNT_X and TT_X along with TNT_U equaling 30% vs. low statistically insignificant TT_U imply that less educated informal workers would gain from working formally in terms of both observable and unobservable characteristics, while highly educated workers more frequently holding formal jobs gain only in terms of observable human capital. This result suggests that access to formal jobs is strongly conditioned on the level of education, and that the allocation of employees is not ruled any more by the sorting on the gain.

The 2006 results thus suggest the relevance of labor market segmentation based on higher education. They imply as well that the inflow of civil workers and young workers into the private sector not only increased informal employment but also decreased the social welfare of paid employment.

When excluding workers with higher education, we found positive sorting on the gain related to the largest extent to observables. Formal/informal wage gaps among the employees without university education are due to the different return to experience and education. The gaps are slightly larger than the corresponding gaps estimated using the whole sample. This may be explained by the stronger formal/informal difference of wage elasticity with respect to experience among the employees without university education than among the whole sample of employees. There is no sorting on the gain based on unobservable characteristics within this population: the corresponding treatment parameters are zero, which is consistent with the MTE graph being flat and close to the abscissa axis (Figure 12). Less educated workers self-select themselves into different employment groups following a higher return on their human capital as long as they do not compete with highly educated workers, where positive sorting becomes impossible.

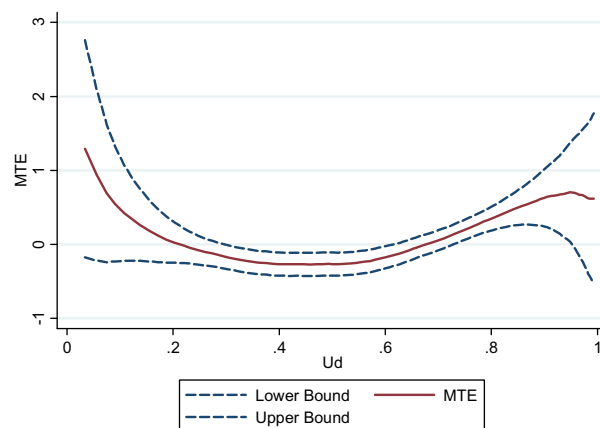


Figure 11. Marginal treatment effect with 95% confidence interval bounds for employees with any level of education, 2006.

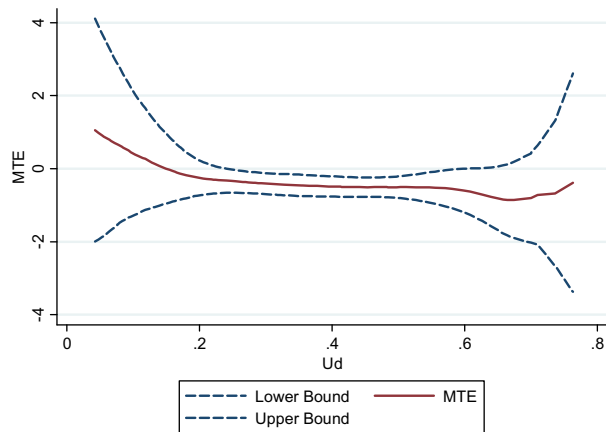


Figure 12. Marginal treatment effect with 95% confidence interval bounds for employees without tertiary education, 2006.

Finally, the selection bias in 2006 is found to be close to zero for the workers without tertiary education. This population is thus quite homogenous across the formal and informal groups in terms of their unobservables.

The differences of the 2006 results based on different samples imply heterogeneity of the mechanisms inducing employees to work informally following labor market restructuring: some

informal employment is voluntary while to the largest extent informal employment is involuntary.

As compared with the 1998 results which show that the nature of employees' allocation into formal *vs.* informal employment is dominantly competitive, 2006 estimates show increasing segmentation of the market and heterogeneity of informal employment. This may be explained by the inflow of civil and young workers in the private sector from 1998 to 2006.

(c) Treatment effects by gender

The central results of the paper discussed in Section 5(b) have been obtained considering the whole population of non-agricultural employees, including both male and female workers. While much less active at the nonagricultural market as compared to men, women participate in both formal and informal employees groups. Their distribution among formal and informal employees is more unequal than that of men (Table 3) in 1998: a majority of female workers (85%) are employed formally. This is related to the high reliance on the public sector until the early 1990s (Assaad, 2005). In 2006, informal female employment rises to 25%.

Given similar informalization trends over the period of study for men and women, we do not consider the labor market separately based on gender. This yields the stronger full support enhancing the validity of the results provided by the

Table 9. ATE, TT, TNT and their decompositions by gender, 1998 and 2006

Treatment effects	1998			2006		
	Total	TT_X ATE_X TNT_X	TT_U ATE_U TNT_U	Total	TT_X ATE_X TNT_X	TT_U ATE_U TNT_U
<i>Female workers with any level of education</i>						
TT	0.238*** (0.023)	1.811	-1.573***	1.022*** (0.101)	0.903	0.119
ATE	-0.088 (-0.088)	1.721	-1.809***	1.051*** (0.118)	0.845	0.206**
TNT	-0.973*** (0.211)	1.214	-2.187***	1.002*** (0.143)	0.659	0.347**
<i>Female workers without university education</i>						
TT	0.201*** (0.087)	1.491	-1.290***	1.269*** (0.125)	1.285	-0.016
ATE	-0.184*** (0.047)	1.345	-1.529***	1.063*** (0.168)	1.137	-0.074
TNT	-0.754*** (0.142)	1.061	-1.815***	0.732*** (0.270)	0.877	-0.145
<i>Male workers with any level of education</i>						
TT	0.255*** (0.047)	1.549	-1.294***	0.324*** (0.112)	0.260	0.064
ATE	-0.406*** (0.072)	1.404	-1.810***	0.417*** (0.125)	0.211	0.206**
TNT	-1.001*** (0.161)	1.111	-2.112***	0.442*** (0.135)	0.142	0.300***
<i>Male workers without university education</i>						
TT	0.304*** (0.105)	1.468	-1.164***	0.398*** (0.127)	0.382	0.016
ATE	-0.235*** (0.058)	1.294	-1.529***	0.183 (0.170)	0.258	-0.075
TNT	-0.754*** (0.105)	1.061	-1.815***	-0.006 (0.257)	0.136	-0.142

Standard Errors are given in parenthesis.

** Significant at the 5% level.

*** Significant at 1%.

model with essential heterogeneity. However, at the level of aggregate treatment effects, the model allows for integrating over individual characteristics for different strata. Table 9 reports the aggregated parameters obtained separately for male and female employees. Not surprisingly, the overall trends and phenomena among different gender groups are similar to the common results discussed in Section 5(b). This is a rather mechanical effect due to the base modeling over the entire population of employees.

However, the separate aggregation yields some interesting gender effects. First, formal/informal wage gap gets particularly strong for women in 2006 (woman's formal wage on average twice as high as her informal wage). This gap is due basically to the observable characteristics and not compensated any more by unobservables. Next, the sorting on the gain in the formal employment (the difference between TT and ATE) that is strong in 1998 for the entire population is weak for women. The latter implies that the sorting on the gain considerations are much less relevant for female employees than for male employees. In line with the descriptive statistics, women have higher propensity for working formally. This may be due to both stronger attachment to job security and low female performance and competitiveness as compared to that of the informal male employees.

6. CONCLUSIONS

This paper brings new methodology to the old discussion on the mechanisms driving informal employment in developing countries. Identifying the right mechanism is crucial for any welfare analysis concerned with informal employment. An assumption on the mechanism ruling informal employment is a starting point of any policy analysis related to the informal sector of the labor market. The competing theories are dual market models and comparative advantage analysis. The empirical evidence presented in the paper offers support to each polar view, implying heterogeneous informal employment. This heterogeneity is related to different reasons that may have different workers for entering the informal sector.

This paper contributes to the debates by offering an empirical study of the Egyptian labor market over 1998–2006, characterized by significant inflows of civil and young workers in the private sector, a steadily increasing level of education, and new legislation favoring formalization through facilitating the hiring/firing process. Thus, the paper provides a case study of one of the lower-middle-income African economies in the context of highly dynamic labor market.

The analysis is carried out using the model with essential heterogeneity recently developed by Heckman et al. (2006). We adjust the estimation procedure proposed by Heckman et al. (2006) to fit the specific context of the sectorial choice qualified by an unequal distribution of the model's observable explanatory variables among two different groups and the discrete nature of these variables. More specifically, we decompose the treatment effects of interest into the effects due to the observables in addition to the effects due to the unobservables. The decomposition is innovative in the framework of the model with essential heterogeneity. In addition, we adjust the construction of the treatment effects for the case of multiple discrete control variables, defining the conditional density of the propensity score associated to the probability of working in formal employment.

The results show triple heterogeneity of formal and informal employees of the Egyptian labor market: heterogeneity of observable and unobservable human capital, heterogeneity of the relative gains or losses from working formally rather than informally, and heterogeneity of the mechanisms inducing employees to work informally.

In 1998, formal paid employment consisted of the higher tier of the employees: employees with high observable and unobservable human capital. Informal employment consisted of the lower-skilled workers. The results imply as well heterogeneity of individual gains due to both observed human capital characteristics and unobserved individual effects. Positive sorting on the gain in 1998 fits the pattern of cost-benefit analysis prevailing as the main mechanism driving informal work in this period. The workers' allocation was efficient from the point of view of relative gains from being employed formally or informally: on average each group of employees benefited most from its actual type of employment.

In 2006, individual heterogeneity between formal and informal employment is reduced: informal employment is no longer a segment for lower-skilled workers; it hosts now highly educated workers and provides returns to education. Heterogeneity of the relative gains from working formally rather than informally has decreased as well: sorting on the gain is found only among employees without tertiary education. Its magnitude is much less important than in 1998, implying small room taken by voluntary informal employment in the allocation process following the market restructuring. The mechanism driving informal work has changed, now consisting to some extent of both a small competitive market component and a large segmentation component.

Overall, the allocation of employees among the formal and informal groups becomes socially inefficient through the deformation process from 1998 to 2006: workers no longer attain the highest relative gains from their actual allocation. Thus, deformation through the public sector reduction decreases not only job security but also the social welfare of workers.

Since the segmentation component becomes the main pattern followed by the Egyptian labor market in 2006, market formalization could not be achieved by improving social packages or introducing a formal minimum wage. An appropriate policy would be tackling the barriers to entry into the formal sector. The Egyptian experience shows that increasing flexibility in regulation of the hiring/firing process is insufficient to balance the inflow of workers into the private sector. Formal jobs became deficient at any level of education. The incentive therefore should be related to formal job creation appealing to various levels of skills. The scarcity of formal jobs at all skill levels combined with the steadily increasing level of education implies that education policy could provide another kind of incentive to inducing young Egyptians to obtain the specific skills required by the sectors of the formal economy which are growing.

Finally, the results imply that policies related to informal employment should address differently various strata, in particular gender groups and groups with different levels of education. Women's engagement in the labor market can be increased through formal employment. Growth of formal employment would reduce gender differences in labor market participation. Women's participation in the labor market not being specifically under the focus of this analysis, further research is needed.

NOTES

1. More detailed theoretical overview can be seen in Kuchera and Roncalato (2008).
2. A third round, ELMPs 12, was not published in time to be included in this study.
3. More specifically, proportional selection of urban/rural primary sampling units (PSU) has been done at first. Then PSUs were randomly selected from the master sample. Twenty-five households were finally selected at random within each of them.
4. Under the market definition of economic activity, only market work counts as work, so that subsistence workers are considered unemployed.
5. Data analysis of agricultural employment evolution could be misleading due to different survey approaches in 1998 and 2006. The share of agriculture in total employment has increased significantly from 20% to 25%. As Assaad (2009) points out, the increased share of agriculture is in part due to measurement issues relating to the involvement of women in agriculture; the bulk of this growth is among unpaid family workers on family farms rather than among hired agricultural workers.
6. Wages are reported in 2006 money. The inflation coefficient is 1.4.
7. The logarithm specification excludes workers with less than 1 year experience. Consequently, the 1998 sample is reduced by about 3.8% (169 workers). The reduction is not, however, harmful for the analysis: only 9 observations of the 2006 sample are excluded. Rather, the specification equilibrates the samples in terms of new entrées.
8. The first evidence is provided by the test proposed in Heckman and Vytlačil (2000): we run regression of wage on polynomial terms of the propensity scores ($p^2, p^3, p^4, \dots$) while controlling for observable wage determinants X and their products with scores $X \cdot p$. We found the coefficients associated with the nonlinear scores terms to be statistically different from zero. The test provides some evidence that $K(p)$ from (7) (Appendix A) is a nonlinear function of p and MTE is heterogeneous among the workers with different propensity scores.

REFERENCES

- Assaad, R. (2009). Labor supply, employment and unemployment in the Egyptian economy, 1988–2006. In R. Assaad (Ed.), *The Egyptian labor market revisited* (pp. 1–52). Cairo: American University in Cairo Press.
- Assaad, R. (2005). Informalization and Defeminization: Explaining the Unusual Pattern in Egypt. In N. Kudva, & L. Beneria (Eds.), *Rethinking informalization: precarious jobs, poverty and social protection* (pp. 86–102). Ithaca, NY: Internet-First University Press.
- Barsoum, G. (2007). Egypt Labor Market Panel Survey 2006: Report on methodology and data collection. *ERF Working Paper 0704*. Cairo: Economic Research Forum.
- Basu, A. (1997). *Analytical development economics*. Cambridge, MA: MIT Press.
- Basu, A., Heckman, J. J., Navarro-Lozano, S., & Urzua, S. (2007). Use of instrumental variables in the presence of heterogeneity and self-selection: An application to treatments in breast cancer patients. *Health Economics*, 16(11), 1133–1157.
- Blundell, R., & Dias, M. C. (2009). Alternative approaches to evaluation in empirical microeconomics. *Journal of Human Resources*, 44(3), 565–640.
- Böhme, M., & Thiele, R. (2012). Is the informal sector constrained from the demand side? Evidence for six West African capitals. *World Development*, 40(7), 1369–1381.
- Carneiro, P., Lokshin, M., Ridao-Cano, C., & Umapathi, N. (2011). Average and marginal returns to upper secondary schooling in Indonesia. *Policy Research Working Paper 5878*. Washington, DC: World Bank.
- Carneiro, P., Heckman, J. J., & Vytlačil, E. J. (2011). Estimating marginal returns to education. *American Economic Review*, 101(6), 2754–2781.
- Cunningham, W. V., & Maloney, W. F. (2002). Heterogeneity in informal self-employment: An application of factor and cluster analysis. *Economic Development and Cultural Change*, 50(1), 131–156.
- Dickens, W. T., & Lang, K. (1985). A test of dual labor market theory. *The American Economic Review*, 67(2), 792–805.
- Dimova, R., Nordman, C., & Roubaud, F. (2010). Allocation of labour in urban west africa. insights from the pattern of labour supply and skill premiums. *Review of Development Economics*, 14(1), 74–92.
- Fields, G. S. (2005a). A welfare economic analysis of labor market policies in the Harris-Todaro model. *Journal of Development Economics*, 76, 127–146.
- Fields, G. S. (2005b). *A guide to multisector labor market models. Social protection discussion paper 0505*. World Bank: Washington, DC.
- Fields, G. S. (1990). Labour market modelling and the urban informal sector: Theory and evidence. In D. Thurnham, B. Salome, & A. Schwarz (Eds.), *The informal sector revisited*. Paris: OECD.
- Floro, M., & Swain, R. (2013). Food security, gender and occupational choice among urban low-income households. *World Development*, 42, 89–99.
- Grimm, M., Krüger, J., & Lay, J. (2011). Barriers to entry and returns to capital in informal activities: Evidence from sub-Saharan Africa. *The Review of Income and Wealth*, 57, 27–53.
- Gunther, I., & Launov, A. (2012). Informal employment in developing countries: Opportunity or last resort?. *Journal of Development Economics*, 97(1), 88–98.
- Harris, J. R., & Todaro, M. P. (1970). Migration, unemployment and development: A two sector analysis. *The American Economic Review*, 60(1), 126–142.
- Heckman, J. J., & Vytlačil, E. J. (2000). Local Instrumental Variables. *NBER Technical Working Papers 0252*, Cambridge, MA: National Bureau of Economic Research, Inc.
- Heckman, J. J., & Hotz, V. (1986). An investigation of the labor market earnings of Panamanian males. *The Journal of Human Resources*, 21(4), 507–542.
- Heckman, J. J., & Li, X. (2004). Selection bias, comparative advantage and heterogeneous returns to education: Evidence from China in 2000. *Pacific Economic Review*, 9(3), 155–171.
- Heckman, J. J., Tobias, J., & Vytlačil, E. J. (2003). Simple estimators for treatment parameters in a latent-variable framework. *Review of Economics and Statistics*, 85(3), 748–755.
- Heckman, J. J., Urzua, S., & Vytlačil, E. J. (2006). Understanding instrumental variables in models with essential heterogeneity. *Review of Economics and Statistics*, 88(3), 389–432.
- Heckman, J. J., & Vytlačil, E. J. (2005). Structural equations, treatment effects and econometric policy evaluation. *Econometrica*, 73(3), 669–738.
- Heckman, J. J., & Vytlačil, E. J. (2007). Econometric evaluation of social programs (1st ed.). In J. J. Heckman, & E. E. Leamer (Eds.), *Handbook of Econometrics* (Vol. 6). Elsevier.
- Husmanns, R. (2004). Measuring the informal economy: From employment in the informal sector to informal employment. *Working Paper 53*. Policy Integration Department, Bureau of Statistics, Geneva: ILO.
- Kuchera, D., & Roncalato, L. (2008). Informal employment: Two contested policy issues. *International Labour Review*, 147(4), 321–348.
- Magnac, T. (1991). Segmented or competitive labor markets. *Econometrica*, 59(1), 165–187.

- Maloney, W. F. (1999). Does informality imply segmentation in urban labor markets? Evidence from sectoral transitions in Mexico. *The World Bank Economic Review*, 13, 275–302.
- Maloney, W. F. (2004). Informality Revisited. *World Development*, 32, 1159–1178.
- Meghir, C., Narita, R., & Robin, J.-M. (2012). Wages and Informality in Developing Countries, *NBER Working Paper* 18347. Cambridge, MA: National Bureau of Economic Research, Inc.
- Navarro-Lozano S., Schrimpf P. (2004). The Importance of Being Formal: Testing for Segmentation in the Mexican Labor Market. Mimeo, University of Chicago.
- Pasquier-Doumer, L. (2013). Intergenerational transmission of self-employed status in the informal sector: A Constrained Choice or Better Income Prospects? Evidence from seven West-African countries. *Journal of African Economies*, 22(1), 73–111.
- Paulson, A. L., & Townsend, R. M. (2005). Financial constraints and entrepreneurship: Evidence from the Thai financial crisis. *Economic Perspectives*, 3Q, 34–48.
- Pratap, S., & Quintin, E. (2006). Are labor markets segmented in developing countries? A semiparametric approach. *European Economic Review*, 50(7), 1817–1841.
- Reyes, L., Rodríguez J., & Urzúa S. S. (2013). Heterogeneous Economic Returns to Postsecondary Degrees: Evidence from Chile. *NBER Working Paper* 1895. Cambridge, MA: National Bureau of Economic Research, Inc.
- Rosenzweig, M. R. (1988). Labor markets in low income countries (1st ed. In H. Chenery, & T. N. Srinivasan (Eds.). *Handbook of development economics* (Vol. 1, pp. 713–762). Elsevier.
- Roy, A. D. (1951). Some thoughts on the distribution of earnings. *Oxford Economic Papers*, 3, 135–146.
- Stiglitz, J. E. (1976). The efficiency wage hypothesis, surplus labor, and the distribution of labour in LDCs. *Oxford Economic Papers*, 28(2), 185–207.
- Wang, X., Fleisher, B. M., Li, H., & Li, S. (2007). Access to Higher Education and Inequality: The Chinese Experiment. *IZA Working Paper*, 2823. Bonn, Germany: Institute for the Study of Labor.

APPENDIX A. ESTIMATION METHOD

The estimation strategy is based on the following representation of model (2)–(4):

$$E(\ln W|X = x, P(Z) = p) = \alpha_0 + \beta_0 x + ((\beta_1 - \beta_0)x)p + K(p) \quad (5)$$

with

$$K(p) = (\alpha_1 - \alpha_0)p + E(U_1 - U_0|P(Z) = p)p + E(U_0|P(Z) = p) \quad (6)$$

MTE then becomes

$$MTE(X, U_D) = \Delta\beta \cdot x + MTE^U$$

where $\Delta\beta = \beta_1 - \beta_0$ and $MTE^U = \frac{\partial K(p)}{\partial p} \Big|_{p=U_D}$. MTE is defined over the common support of distributions of the propensity scores predicted for two groups, formal and informal. The aggregated treatment effects are identified over the full support (common support covering whole the population under consideration).

The estimation procedure implies several steps:

- (1) We first estimate the propensity scores (predicted probabilities for holding job 1) using probit estimation of selection model (4).

As shown in Heckman and Vytlačil (2005) and Heckman et al. (2006), one of the assumptions underlying the model identification is the nondegenerate distribution of I conditional on X . It implies an exclusion variable impacting the

probability to participate in the formal sector but not impacting the wage. An instrument available in our data set is a dummy variable indicating for each worker whether another member of the household participates in the formal sector. As Table 5 indicates, there is a relatively strong significant impact of this variable on the variation of the propensity scores. On the other hand, the residuals of Eq. (7) estimated using Eq. (8) (see (3) and (4) below) and representing the residuals of the model wage equation are not correlated with the dummy.

- (2) We construct then the common support of the propensity distributions for workers from both sectors: $(\max(p_{0,\min}, p_{1,\min}); \min(p_{0,\max}, p_{1,\max}))$, where $p_{j,\min}$, $p_{j,\max}$ are the minimal and maximal values of propensity scores for population $D = j$ ($j = 0, 1$).

- (3) Next, β_0 and $(\beta_1 - \beta_0)$ are estimated by the double residual semi-parametric regression of Eq. (7) associated with (5)–(6):

$$\ln W = \alpha_0 + \beta_0 X + ((\beta_1 - \beta_0)X) \cdot p + K(p) + v \quad (7)$$

where $K(p)$ is a nonparametric function of p and $E(v|X, p) = 0$.

We use Epanechnikov density when calculating kernels and run estimation at the common support points.

- (4) The nonparametric term $K(p)$ is associated then with the residual $\tilde{Y}$ of Eq. (7):

$$\tilde{Y} = \ln W - \hat{\beta}_0 \cdot X - ((\hat{\beta}_1 - \hat{\beta}_0) \cdot X) \cdot p$$

$$K(p) + \alpha_0 = E(\tilde{Y}|p) \quad (8)$$

$\frac{\partial K(p)}{\partial p}$ is obtained thus by running local quadratic regression of $\tilde{Y}$ on p . The estimator of the first derivative is given by the linear term coefficient of the local polynomial regression. We chose thus to increase the polynomial degree by one as recommended in Blundell and Dias (2009) and apply the quadratic regression. We use Epanechnikov density when calculating kernels and estimation grid made up of the common support centiles.

- (5) The average conditional effects $ATE(x)$, $TT(x)$, $TNT(x)$ are calculated as MTE weighted means:

$$ATE(x) = \int_0^1 MTE(x, u_D) du_D = \Delta\beta x + \int_0^1 MTE^U du_D$$

$$TT(x) = \int_0^1 MTE(x, u_D) w_{TT}(x, u_D) du_D = \Delta\beta x + \int_0^1 MTE^U w_{TT}(x, u_D) du_D$$

$$TNT(x) = \int_0^1 MTE(x, u_D) w_{TNT}(x, u_D) du_D = \Delta\beta x + \int_0^1 MTE^U w_{TNT}(x, u_D) du_D$$

where $w_{TT}(x, u_D)$ and $w_{TNT}(x, u_D)$ weight stronger the observations with higher probability of holding job 1 or job 0 respectively:

$$w_{TT}(x, u_D) = \frac{\Pr(P(Z) > u_D | X = x)}{\int \Pr(P(Z) > u_D | X = x) du_D}$$

$$w_{TNT}(x, u_D) = \frac{\Pr(P(Z) < u_D | X = x)}{\int \Pr(P(Z) < u_D | X = x) du_D}$$

The MTE being a function of both observables X and unobservables U_D , their joint distribution is formalized by $\Pr(P(Z) > u_D | X = x)$, $\Pr(P(Z) < u_D | X = x)$. Most of empirical works using the model with essential heterogeneity, proceed through aggregating X (which is a high dimensional object) into $(\beta_1 - \beta_0) \cdot X$ or predicting the conditional density using the Probit model applied to the $I(P(Z) > u_D)$ (Heckman et al. (2006), Carneiro, Heckman et al. (2011), Carneiro, Lokshin et al. (2011b)). The first approach is criticized by Carneiro, Heckman et al. (2011), Carneiro, Lokshin et al. (2011) as having little theoretical basis. The second approach suffers as well from a theoretical constraint implying a specific form of the distribution (the normal one when using Probit). In addition, it is not feasible in our case where the explanatory variables are discrete (dummy in general). Indeed, a resampling necessary to bootstrap the estimation (see point 7) implies outcomes predicting perfectly the index $I(P(Z) > u_D)$ or inducing too weak variation of outcomes, implying unfeasible maximization of the likelihood function. We adopt thus another procedure using the empirical distribution of the propensity scores conditional on the joint distribution of X affecting differently the wage in the informal and formal sectors. Indeed, the joint distribution of the discrete variables implies a limited number of groups. $I(P(Z) > u_D)$ is easily calculated then within each group using the mean value of I .

(6) Unconditional effects (population average effects ATE , TT , TNT) are found by integrating X out.

$$ATE = ATE_U + ATE_X \quad (9)$$

$$\text{with } ATE_X = \int \Delta \hat{\beta} \cdot x dF_X(x) \quad \text{and} \quad ATE_U = \int \int_0^1 MTE^U du_D dF_X(x).$$

$$TT = TT_X + TT_U \quad (10)$$

$$\text{with } TT_X = \int \Delta \hat{\beta} \cdot x dF_{X|D=1}(x) \quad \text{and} \quad TT_U = \int \int_0^1 MTE^U w_{TT}(x, u_D) du_D dF_{X|D=1}(x).$$

$$TNT = TNT_X + TNT_U \quad (11)$$

$$\text{with } TNT_X = \int \Delta \hat{\beta} \cdot x dF_{X|D=0}(x) \quad \text{and} \quad TNT_U = \int \int_0^1 MTE^U w_{TT}(x, u_D) du_D dF_{X|D=0}(x)$$

Thus, formal/informal wage gaps are decomposed into heterogeneous returns to the observable and unobservable wage determinants.

(7) Statistical inference about estimators (ATE , TT , TUT variance and t -statistics; MTE 95% confidence interval) is obtained by using a bootstrap technique.

APPENDIX B. OLS ESTIMATES

Tables 10 and 11.

Table 10. 1998 wage equation: OLS estimates, workers with any level of education

Worker's demographic characteristic	1998, Return from informal employment $\hat{\beta}_0(\text{SE})$	1998, Gap in formal-informal return $(\hat{\beta}_1 - \hat{\beta}_0)(\text{SE})$	2006, Return from informal employment $\hat{\beta}_0(\text{SE})$	2006, Gap in formal-informal return $(\hat{\beta}_1 - \hat{\beta}_0)(\text{SE})$
Constant	-0.282** (0.073)	-0.333*** (0.090)	-0.207*** (0.066)	-0.028 (0.090)
Ln(experience)	0.206*** (0.020)	0.091*** (0.023)	0.168*** (0.019)	0.118*** (0.023)
Primary education	0.106** (0.047)	0.056 (0.065)	0.017 (0.048)	0.165** (0.073)
Preparatory education	0.112* (0.063)	0.092 (0.083)	0.091 (0.066)	0.189** (0.092)
Intermediate education	0.175*** (0.047)	0.273*** (0.059)	0.087** (0.041)	0.427*** (0.057)
Tertiary/university education	0.263*** (0.064)	0.522*** (0.073)	0.223*** (0.053)	0.594*** (0.066)
Female	-0.282*** (0.057)	0.168*** (0.063)	-0.397*** (0.044)	0.458*** (0.051)
Greater Cairo (reference group)	—	—	—	—
Alexandria and Suez Canal Cities	-0.030** (0.030)	—	-0.048 (0.032)	—
Urban Lower Egypt	-0.143*** (0.028)	—	-0.148*** (0.032)	—
Urban Upper Egypt	-0.154*** (0.028)	—	-0.092*** (0.029)	—
Rural Lower Egypt	-0.208*** (0.029)	—	-0.144*** (0.029)	—
Rural Upper Egypt	-0.189*** (0.034)	—	-0.129*** (0.033)	—
Marital status	0.063*** (0.024)	—	0.102*** (0.024)	—
Number of observations	4,002	6,847		

* Significant at the 10% level;

** Significant at the 5% level;

*** Significant at the 1% level.

Table 11. 1998 wage equation: OLS estimates, workers without university education

Worker's demographic characteristic	1998, Return from informal employment $\hat{\beta}_0(\text{SE})$	1998, Gap in formal–informal return $(\hat{\beta}_1 - \hat{\beta}_0)(\text{SE})$	2006, Return from informal employment $\hat{\beta}_0(\text{SE})$	2006, Gap in formal–informal return $(\hat{\beta}_1 - \hat{\beta}_0)(\text{SE})$
Constant	−0.371*** (0.080)	−0.209** (0.111)	−0.240*** (0.072)	−0.068 (0.109)
Ln(experience)	0.201*** (0.022)	0.072** (0.030)	0.170*** (0.022)	0.151*** (0.029)
Primary education	0.103*** (0.047)	0.057 (0.066)	0.016 (0.046)	0.177** (0.072)
Preparatory education	0.119* (0.064)	0.081 (0.084)	0.096 (0.063)	0.203** (0.090)
Intermediate education	0.185*** (0.049)	0.215*** (0.062)	0.087** (0.040)	0.408*** (0.058)
Female	−0.331*** (0.064)	−0.238*** (0.080)	−0.389*** (0.051)	−0.532*** (0.065)
Greater Cairo (reference group)	—	—	—	—
Alexandria and Suez Canal Cities	0.039 (0.039)	—	0.013 (0.041)	—
Urban Lower Egypt	−0.069* (0.037)	—	−0.084** (0.040)	—
Urban Upper Egypt	−0.152*** (0.037)	—	−0.075** (0.038)	—
Rural Lower Egypt	−0.140*** (0.035)	—	−0.095*** (0.035)	—
Rural Upper Egypt	−0.135*** (0.040)	—	−0.076** (0.038)	—
Marital status	0.094*** (0.030)	—	0.089*** (0.029)	—
Number of observations	4,002	6,822		

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

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